Project 3 Harmonic Analysis

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1 Introduction

Definition 1 (Weights)**.** *A* weight *is a function $w \in L^1_{\mathrm{loc}}(\mathbb{R}^d)$ with $w \geq 0$. For a measurable set $E \subseteq \mathbb{R}^d$ we write $w(E) := \int_E w(x)\,dx$.*

Definition 2 (Muckenhoupt A_p weights)**.** *Let $1 < p < \infty$ and let w be a weight. The A_p* characteristic *of w is*

$$[w]_{A_p} := \sup_Q \left(\frac{1}{|Q|}\int_Q w(x)\,dx\right)\left(\frac{1}{|Q|}\int_Q w(x)^{-\frac{1}{p-1}}\,dx\right)^{p-1}, \tag{1}$$

where the supremum is taken over all cubes $Q \subset \mathbb{R}^d$ with sides parallel to the axes. We say that $w \in A_p$ if $[w]_{A_p} < \infty$.

Definition 3 (Hardy-Littlewood maximal operator with respect to w)**.** *Let w be a weight. For a measurable function f and $x \in \mathbb{R}^d$,*

$$M_w f(x) := \sup_{Q \ni x} \frac{1}{w(Q)}\int_Q |f(y)|\,w(y)\,dy, \tag{2}$$

where the supremum is taken over all cubes Q containing x. If the supremum is taken only over cubes Q centred at x, we denote the corresponding operator by M^c_w. When $w \equiv 1$ we drop the subscript and write M and M^c.